terminal *A* becomes positive with respect to terminal *B*. The diode is forward biased and hence it conducts. The current flows through the load resistor R_L and the AC voltage developed across R_L constitutes the output voltage V_0 and the waveform of the output voltage is shown in Figure 10.17(b).

During the negative half cycle

When the negative half cycle of the AC input signal passes through the circuit, terminal *A* is negative with respect to terminal *B*. Now the diode is reverse biased and does not conduct. Hence no current passes through R_L. The reverse saturation current in a diode is negligible. Since there is no voltage drop across R_L, the negative half cycle of AC supply is suppressed at the output.

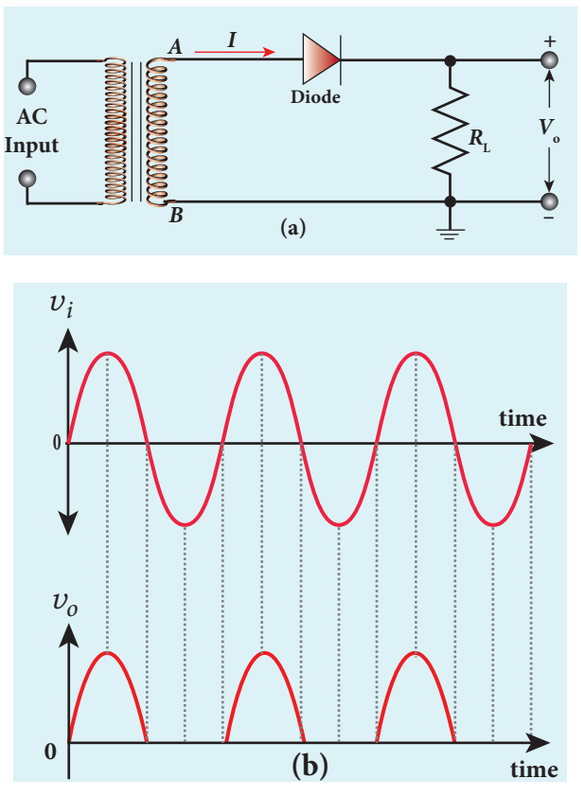


Figure 10.17 (a) Half wave rectifier circuit (b) Input and output waveforms

The output of the half wave rectifier is not a steady DC voltage but a pulsating wave. This pulsating voltage cannot be used for electronic equipments. A constant or a steady voltage is required which can be obtained with the help of filter circuits and voltage regulator circuits.

Efficiency (η) is the ratio of the output DC power to the AC input power supplied to the circuit. Its value for half wave rectifier is 40.6 %.

Note If the direction of the diode is reversed, the negative half of the AC signal is passed through and the positive half is blocked.

ii) Full wave rectifier

The positive and negative half cycles of the AC input signal are rectified in this circuit and hence it is called the full wave rectifier. The circuit is shown in Figure 10.18(a). It consists of two *p-n* junction diodes, a centre tap transformer and a load resistor R_L. The centre is usually taken as the ground or zero voltage reference point. With the help of the centre tap transformer, each diode rectifies one half of the total secondary voltage.

During positive half cycle

When the positive half cycle of the AC input signal passes through the circuit, terminal *M* is positive, *C* is at zero potential and *N* is at negative potential. This forward biases diode D_1 and reverse biases diode D_2. Hence, being forward biased, diode D_1 conducts and current flows along the path MD_1ABC.

During negative half cycle

When the negative half cycle of the AC input signal passes through the circuit, terminal *N* becomes positive, *C* is at zero

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